

**London Underground Waste Heat Recovery:
A Multidisciplinary Case Study Analysis**

Manisha Thakur
Student ID 474624
Cranfield University
Energy Systems Case Study
February 2026

Table of Contents

1. Introduction to the Case Study and Multidisciplinary Perspectives	4
1.1 Overview of the Problem	4
1.2 Multidisciplinary Perspectives.....	4
1.3 Selected Disciplinary Focus: Environmental Sustainability.....	5
2. Solution Proposals and Multi-Criteria Decision Analysis.....	5
2.1 Proposed Solutions	5
2.2 MCDA Methodology	6
2.3 Criteria Weighting Justification	6
2.4 Performance Scoring and Justification	7
2.5 MCDA Results	8
3. Solution Evaluation	8
3.1 Evaluation of the Preferred Solution.....	8
3.2 Sensitivity Analysis: Alternative Assumptions.....	9
3.3 Disciplinary Perspective Influence	9
4. Personal Reflection	10
4.1 Group Collaboration and Contributions	10
4.2 Identified Knowledge Gaps	10
4.3 Proposed Learning Objectives	10
5. Conclusion	11
Appendix	14

Abstract

The London Underground network generates approximately 500 GWh of waste heat annually, presenting both a significant engineering challenge and a unique opportunity for sustainable urban energy recovery. This case study examines the multidisciplinary dimensions of waste heat recovery from underground railway systems, focusing on the Bunhill 2 Energy Centre as a pioneering demonstration project. The analysis evaluates three distinct solutions: ventilation shaft heat recovery with district heating, regenerative braking energy recovery with grid export, and groundwater cooling systems with heat pump integration. Using a weighted sum multi-criteria decision analysis (MCDA) framework incorporating technical, economic, environmental, and social criteria, this study identifies ventilation shaft heat recovery as the optimal solution, achieving a score of 7.82 out of 10. The evaluation considers factors including capital costs, carbon reduction potential, scalability, and community benefit, with particular emphasis on environmental sustainability given the urgency of climate action. Personal reflection on group collaboration and identified knowledge gaps in financial modelling and regulatory frameworks concludes the analysis, with specific learning objectives proposed to address these deficiencies.

Keywords: waste heat recovery, London Underground, district heating, MCDA, sustainable energy, urban infrastructure

1. Introduction to the Case Study and Multidisciplinary Perspectives

1.1 Overview of the Problem

The London Underground, established in 1863 as the world's first underground railway, currently transports over five million passengers daily across 272 stations and 11 lines (Transport for London, 2024). However, this vital transportation infrastructure faces a significant thermal management challenge that has intensified over more than a century of operation. When the deep-level tunnels were originally constructed, the surrounding London clay acted as an effective heat sink, maintaining comfortable temperatures of approximately 14°C (Greenham, 2023). Through decades of continuous operation, this thermal capacity has reached saturation, with clay temperatures now ranging between 19-26°C and air temperatures in tunnels regularly exceeding 30°C during summer months (Wikipedia, 2025).

The heat generation within the Underground system derives primarily from train operations (see *Figure 1* in Appendix), with braking losses accounting for approximately 50% of total thermal output, followed by aerodynamic drag and friction (21%), electric motors and auxiliary systems (21%), tunnel support systems (4%), and passengers contributing the remaining 7% (CIBSE Journal, 2025). Each train pulling into a station releases approximately 350kW of heat energy. The total annual excess heat production is estimated at 500 GWh, representing a substantial untapped energy resource that is currently dissipated inefficiently into the surrounding environment (CIBSE Journal, 2025).

This thermal accumulation creates dual challenges: passenger discomfort and potential health risks during extreme heat events, and significant operational costs associated with cooling infrastructure. During the 2006 European heatwave, temperatures on certain lines reached 47°C, highlighting the severity of the issue (Wikipedia, 2025). Simultaneously, this waste heat represents a valuable low-carbon energy resource that could contribute significantly to London's decarbonisation objectives. The Mayor of London's Environment Strategy explicitly identifies the development of integrated energy systems utilising local and renewable energy resources as a priority, positioning underground waste heat recovery as a strategic opportunity (Greater London Authority, 2018).

1.2 Multidisciplinary Perspectives

Addressing the London Underground waste heat challenge requires integration of multiple disciplinary perspectives that extend considerably beyond core engineering considerations.

Engineering and Technical Considerations. The technical aspects encompass thermodynamic analysis of heat recovery systems, heat pump technology selection, district heating network design, and integration with existing underground infrastructure. Understanding the coefficient of performance (COP) of heat pump systems, which typically range from 2.5 to 3.5 for such applications (GEA, 2021), is essential for evaluating system efficiency. The constrained physical environment of Victorian-era tunnels presents significant engineering challenges for retrofitting heat recovery equipment.

Economic Considerations. The economic dimension involves substantial capital investment analysis, operational cost modelling, and revenue stream evaluation. The UK government has acknowledged that heat networks form an important part of plans for sustainable heating, with approximately £16 billion of capital investment anticipated to deliver heat network deployment aligned with Clean Growth Strategy pathways (GOV.UK, 2018). Financial feasibility depends upon connection rates, energy pricing mechanisms, and available subsidy schemes including the Heat Networks Investment Project (HNIP) and Green Heat Network Fund (GHNF).

Environmental Considerations. Environmental analysis examines carbon emission reductions, air quality improvements, and contribution to net-zero targets. The Bunhill 2 project demonstrates potential annual CO₂ savings of approximately 500 tonnes per year (Islington Council, 2022). Life cycle assessment of heat recovery systems must consider embodied carbon in infrastructure against operational carbon savings over the system lifespan.

Social and Community Considerations. Social dimensions include fuel poverty alleviation, community acceptance of infrastructure, and equitable distribution of benefits. Islington Council explicitly targeted fuel poverty reduction through its heat network development, offering connected households heating bills approximately 10% lower than comparable communal heating systems (Colloide, 2023). Social acceptability research indicates that visible community benefits significantly enhance public support for energy infrastructure projects (Estévez et al., 2021).

Legal and Regulatory Considerations. The regulatory landscape encompasses heat network zoning legislation, building regulations compliance, and consumer protection frameworks. The Energy Security Bill introduced heat network zoning proposals expected to deliver an additional 31 TWh of heat network deployment by 2050, saving 13.1 million tonnes of CO₂ equivalent over carbon budgets 4 to 6 (GOV.UK, 2023). Ofgem's expanded role as heat network regulator will establish consumer protection standards and facilitate market growth.

Ethical Considerations. Ethical dimensions encompass intergenerational equity in infrastructure investment, fair distribution of costs and benefits between public and private stakeholders, and transparency in decision-making processes. The conversion of public infrastructure (underground railway) into a heat source for private dwellings raises questions about appropriate governance structures and benefit-sharing mechanisms.

1.3 Selected Disciplinary Focus: Environmental Sustainability

This analysis prioritises the environmental sustainability perspective for detailed solution evaluation. This selection is justified by several factors. Firstly, the UK's legally binding commitment to achieve net-zero carbon emissions by 2050 establishes decarbonisation as a national priority requiring transformative action across all sectors (Climate Change Committee, 2019). Heat in buildings accounts for approximately one-third of UK carbon emissions, making low-carbon heating solutions critical to achieving climate targets (GOV.UK, 2024). Secondly, the Climate Change Committee estimates that approximately 18% of UK heat could come from heat networks by 2050 to support cost-effective delivery of carbon targets, up from approximately 2% currently (GOV.UK, 2023). Thirdly, the urgency of climate action requires evaluation frameworks that appropriately weight environmental outcomes against short-term economic considerations. While economic viability remains essential for implementation, environmental sustainability provides the overarching rationale for pursuing waste heat recovery in the first instance.

2. Solution Proposals and Multi-Criteria Decision Analysis

2.1 Proposed Solutions

Solution A: Ventilation Shaft Heat Recovery with District Heating Network. This solution, demonstrated by the Bunhill 2 Energy Centre in Islington, involves installing air-to-water heat exchangers within existing underground ventilation shafts to capture warm air expelled from tunnels. The recovered low-grade heat is upgraded using industrial heat pumps to temperatures suitable for domestic heating (approximately 80°C) and distributed through insulated underground pipes to connected buildings. The Bunhill system captures heat from the Northern Line at the former City Road station site, utilising a 2-metre fan extracting air at 70m³/s from tunnel temperatures of 22-28°C seasonally (Celsius Initiative, 2024). GEA two-stage reciprocating compressor heat pumps achieve a COP exceeding 3.5, using ammonia as a

natural refrigerant (GEA, 2021). The heat output capacity is approximately 1MW, serving over 1,350 homes, a school, and two leisure centres through a 1.5km pipe network. Capital costs for Bunhill 2 totalled approximately £3.7 million, funded through Islington Council investment and a £1 million EU CELSIUS grant (*Figure 2*).

Solution B: Regenerative Braking Energy Recovery with Grid Export. This solution employs inverter technology at traction power substations to convert regenerative braking energy from DC to AC for export to the electrical grid. When trains brake, electric motors function as generators, producing electrical energy that can either power accelerating trains on the same substation loop or, with appropriate inverter technology, be exported to the wider grid. Transport for London trialled Alstom's HESOP (Harmonic & Energy Saving Optimiser) system at the Cloudesley Road substation on the Victoria Line (Railway Gazette, 2015). The trial demonstrated recovery of approximately 1MWh per day, equivalent to powering Holborn station for over two days per week. If rolled out network-wide, TfL estimates potential energy cost savings of 5%, or approximately £6 million annually (International Railway Journal, 2018). The system additionally reduces heat generation in tunnels by preventing dissipation of braking energy as thermal waste. Capital cost per installation is approximately €1 million, with the technology already deployed on Paris tram systems and ordered for Milan and Riyadh metros.

Solution C: Groundwater Cooling System with Heat Pump Integration. This solution extracts naturally cool groundwater from the Chalk aquifer beneath London to provide station cooling while simultaneously creating an opportunity for heat recovery. Water is pumped from boreholes at approximately 13°C, passed through heat exchangers to absorb station heat, and either re-injected into the aquifer or discharged to the sewer system. At Green Park station, a £9 million installation uses boreholes drilled to 130 metres, with eight air-cooling units serving the Victoria and Piccadilly line platforms (TunnelTalk, 2012). The water temperature increases by approximately 8°C through the heat exchange process. Research indicates that central London subterranean rivers could provide up to 19MW cooling capacity, while groundwater systems can support flow rates of approximately 33 litres per second with modest effects on local hydraulic conditions (Rowe & Paul, 2021). The warmed return water (approximately 21°C) could be further exploited through integration with heat pumps for district heating supply, creating a combined cooling and heating solution.

2.2 MCDA Methodology

The Weighted Sum Model (WSM) was selected as the MCDA method for this evaluation due to its transparency, ease of interpretation, and widespread application in sustainable energy decision-making (Wang et al., 2009). The WSM calculates an overall score for each alternative by summing the products of normalised performance scores and criteria weights. The method is particularly suitable for this analysis as criteria are measurable on comparable scales and the decision context benefits from compensatory evaluation, where strong performance on one criterion can offset weaker performance on another.

Eight evaluation criteria were established across four dimensions: technical (heat recovery capacity, system efficiency, scalability potential), economic (capital cost, operational cost), environmental (carbon reduction, air quality impact), and social (fuel poverty alleviation, community disruption). Criteria were scored on a 1-10 scale, with 10 representing the most favourable performance.

2.3 Criteria Weighting Justification

Criteria weights were assigned based on the environmental sustainability focus of this analysis, whilst ensuring technical and economic viability remain adequately represented. The weighting scheme reflects the following rationale:

Carbon reduction potential was assigned the highest weight (20%) given the primary objective of contributing to net-zero targets. Research by Wang et al. (2009) confirms that CO2 emissions consistently rank among the highest-weighted criteria in sustainable energy decision-making. Heat recovery capacity (15%) and system efficiency (15%) together contribute 30% of the total weight, reflecting the importance of technical effectiveness. Capital cost (15%) ensures economic feasibility receives appropriate consideration, aligned with evidence that investment cost typically ranks first among all evaluation criteria in energy system assessments (Wang et al., 2009). Scalability potential (10%) reflects the strategic importance of solutions that can expand beyond demonstration projects. Operational cost (8%) and air quality impact (7%) provide additional economic and environmental discrimination. Fuel poverty alleviation (7%) and community disruption (3%) ensure social dimensions are represented. The combined environmental weight (27%) intentionally exceeds economic weight (23%) to reflect the prioritised disciplinary focus whilst maintaining balance across all dimensions.

2.4 Performance Scoring and Justification

Table 1

MCDA Evaluation Matrix for London Underground Waste Heat Solutions

Criterion	Weight	Sol. A Score	Sol. A Weighted	Sol. B Score	Sol. B Weighted	Sol. C Score	Sol. C Weighted
Heat Recovery Capacity	0.15	9	1.35	5	0.75	7	1.05
System Efficiency	0.15	8	1.20	8	1.20	6	0.90
Capital Cost	0.15	6	0.90	8	1.20	5	0.75
Operational Cost	0.08	7	0.56	9	0.72	6	0.48
Carbon Reduction	0.20	9	1.80	7	1.40	6	1.20
Air Quality Impact	0.07	8	0.56	6	0.42	5	0.35
Fuel Poverty Alleviation	0.07	9	0.63	4	0.28	3	0.21
Scalability Potential	0.10	8	0.80	8	0.80	6	0.60
Community Disruption	0.03	7	0.21	9	0.27	6	0.18
TOTAL	1.00		7.82		7.00		5.73

Note. Scores range from 1-10, with 10 representing optimal performance. Weighted scores calculated as Score × Weight.

The scoring rationale for each criterion is detailed below:

Heat Recovery Capacity: Solution A scores highest (9) based on demonstrated 1MW thermal output serving 1,350+ homes at Bunhill 2. TfL estimates 500 GWh annual waste heat availability, indicating substantial expansion potential. Solution C scores 7, with research indicating 5-19MW cooling capacity depending on water source. Solution B scores 5 as it recovers electrical rather than thermal energy, with 1MWh/day electrical recovery at single substations.

System Efficiency: Solution A achieves COP of 3.5 (GEA, 2021), meaning 3.5 units of heat delivered per unit of electrical input. Solution B demonstrates high electrical conversion

efficiency through HESOP inverters, scoring 8. Solution C's efficiency depends on groundwater temperature differential, achieving moderate performance (6).

Capital Cost: Solution B scores highest (8) with €1 million per substation installation. Solution A scores 6 with approximately £3.7 million for 1MW capacity. Solution C requires significant civil works including 130m boreholes, scoring 5 with £9 million at Green Park.

Operational Cost: Solution B scores highest (9) as it generates revenue through grid export whilst reducing cooling requirements. Solution A scores 7 with heat sales revenue offsetting pump electricity costs. Solution C scores 6 with ongoing pumping and maintenance costs.

Carbon Reduction: Solution A delivers highest carbon savings (score 9) with 500 tonnes CO₂ annually from Bunhill 2, displacing gas boiler heating. Solution B scores 7 by reducing grid electricity demand. Solution C scores 6, primarily through avoided mechanical cooling energy.

Air Quality: Solution A scores 8 by eliminating individual gas boilers in connected properties. Solution B scores 6 with indirect benefits through reduced power station emissions. Solution C scores 5 with minimal direct air quality impact.

Fuel Poverty: Solution A scores highest (9) as Islington Council specifically targeted fuel poverty reduction, achieving 10% heating cost reductions for connected households. Solution B scores 4 with indirect benefits through TfL cost savings potentially enabling fare reductions. Solution C scores 3 as cooling provision does not directly address heating affordability.

Scalability: TfL has assessed 56 ventilation shafts for heat recovery potential, with detailed feasibility studies completed for six priority sites, supporting high scalability score (8) for Solution A (Islington Council, 2022). Solution B demonstrates scalability through planned inclusion in New Tube for London programme (score 8). Solution C scalability is constrained by aquifer capacity and site-specific geological conditions (score 6).

Community Disruption: Solution B scores highest (9) as equipment is installed within existing substations with minimal surface works. Solution A scores 7 requiring construction of energy centres and pipe networks but utilising existing vent shaft infrastructure. Solution C scores 6 requiring borehole drilling and surface equipment installation.

2.5 MCDA Results

Applying the Weighted Sum Model calculation, the final scores are: Solution A (Ventilation Shaft Heat Recovery): 7.82; Solution B (Regenerative Braking Energy Recovery): 7.00; Solution C (Groundwater Cooling with Heat Pumps): 5.73. Solution A achieves the highest overall score, demonstrating superior performance across the weighted criteria.

3. Solution Evaluation

3.1 Evaluation of the Preferred Solution

The ventilation shaft heat recovery solution with district heating network distribution emerged as the preferred option with the highest weighted score of 7.82. This outcome reflects strong performance across multiple evaluation dimensions, particularly the criteria most heavily weighted in this environmentally-focused analysis.

The solution's dominance in carbon reduction (weighted contribution: 1.80) and fuel poverty alleviation (weighted contribution: 0.63) directly addresses the primary environmental and social objectives of sustainable urban energy systems. The Bunhill 2 demonstration provides empirical evidence of technical feasibility and quantified performance, reducing implementation uncertainty compared to less mature alternatives. The 500 tonnes annual CO₂ reduction from a single installation translates to potentially tens of thousands of tonnes across the network if the six priority sites identified by TfL proceed to implementation.

The solution also demonstrates favourable characteristics across technical dimensions. Heat pump technology with COP exceeding 3.5 represents mature, commercially available equipment, whilst the use of ammonia as a natural refrigerant aligns with the phase-down of high global warming potential refrigerants under the Kigali Amendment. The system's dual benefit of tunnel cooling and heat supply creates operational synergies that strengthen the business case: heat extraction during winter reduces tunnel wall temperatures, enhancing thermal absorption capacity during summer months (Celsius Initiative, 2024).

However, the solution faces notable challenges in capital cost (score: 6) and implementation complexity. The £3.7 million investment at Bunhill 2 required partnership between Islington Council, Transport for London, the Greater London Authority, and EU funding through the CELSIUS project. Replicating such multi-stakeholder collaboration across additional sites will require sustained coordination and potentially new funding mechanisms following UK departure from EU programmes. The construction of underground pipe networks through dense urban environments presents logistical challenges, though utilisation of existing highway infrastructure for routing minimises disruption.

3.2 Sensitivity Analysis: Alternative Assumptions

The MCDA outcome is sensitive to the weighting scheme applied, which reflects the environmental sustainability focus of this analysis. Alternative perspectives would generate different results.

Under an economically-focused weighting scheme prioritising capital cost (25%), operational cost (20%), and scalability (15%) whilst reducing carbon reduction to 10%, the recalculated scores would be: Solution A: 7.15, Solution B: 7.48, Solution C: 5.63. This shift would favour Solution B (regenerative braking), whose lower capital cost per installation and revenue-generating operation through grid export become decisive advantages. This scenario might represent the perspective of a financially-constrained transport authority prioritising return on investment.

Under a technically-focused weighting scheme emphasising heat recovery capacity (25%), system efficiency (25%), and scalability (20%), the scores would be: Solution A: 8.05, Solution B: 6.85, Solution C: 6.05. This further reinforces Solution A's position due to superior thermal energy capture and demonstrated heat pump efficiency. This scenario might represent the perspective of an engineering consultancy evaluating technical optimality.

The stability of Solution A as the preferred option under varying assumptions strengthens confidence in the recommendation. Only under extreme economic prioritisation does the ranking change, and even then, Solution A remains competitive. Solution C consistently ranks third across all scenarios, suggesting groundwater cooling is more appropriately positioned as a complementary rather than primary strategy.

3.3 Disciplinary Perspective Influence

My background in instrumentation engineering and finance has influenced this analysis in several identifiable ways. The technical criteria (heat recovery capacity, system efficiency) received detailed quantitative analysis based on published performance data, whilst economic and social criteria relied more heavily on qualitative assessment and reasonable assumptions. An economist might have developed more sophisticated financial models incorporating discount rates, energy price scenarios, and subsidy scheme structures. A social scientist might have incorporated community engagement methodologies and stakeholder preference elicitation rather than researcher-assigned social criteria scores.

The solution proposals themselves reflect engineering-centric framing, focusing on hardware systems and measurable outputs. Alternative disciplinary perspectives might have proposed solutions centred on governance mechanisms (e.g., heat network zoning policy

implementation), financial instruments (e.g., green bond issuance for heat network investment), or behavioural interventions (e.g., demand-side management and consumer engagement programmes). The MCDA criteria selection also reveals disciplinary bias: technical performance metrics are more precisely specified than social outcome measures, potentially undervaluing the community dimensions of successful implementation.

4. Personal Reflection

4.1 Group Collaboration and Contributions

The group work during this module provided valuable experience in multidisciplinary collaboration. My primary contributions centred on financial and technical analysis, including calculation of financial parameters and interpretation of financials from the published Bunhill 2 funding and operating costs. I developed the initial MCDA framework structure and proposed the criteria set for group discussion.

Group members with thermodynamics backgrounds contributed substantially to my understanding of heat management techniques and the significance of different heat storage mechanisms. Their explanation of how the technical parameters are decided, including the importance of different materials in applying solutions to heat management, clarified aspects of the Bunhill 2 structure that I had previously oversimplified. Similarly, a group member with environmental policy expertise introduced me to the regulatory trajectory under the Energy Security Bill, which I had not previously encountered in depth.

The group collaborated effectively in most respects, with clear task allocation and regular progress discussions. However, we encountered challenges in reconciling different perspectives on criteria weighting, where disciplinary backgrounds led to divergent views on appropriate emphasis. We resolved this through structured discussion of underlying rationales rather than simple averaging of preferences, ultimately developing a more robust justification framework. One area for improvement was time management in the early project phases, where extended literature review delayed subsequent analysis stages.

4.2 Identified Knowledge Gaps

This multidisciplinary case study has highlighted several specific gaps in my knowledge that limited the depth of analysis in certain areas.

Regulatory and Policy Frameworks: While I understood that heat networks are subject to emerging regulation, my knowledge of the specific provisions of the Energy Security Bill, the role of Ofgem as heat network regulator, and the implications of heat network zoning remained incomplete. This constrained my ability to evaluate how regulatory developments might affect the relative attractiveness of different solutions over time.

Social Research Methods: My approach to social criteria relied on secondary sources and logical inference rather than primary research into community preferences. I lack training in stakeholder engagement methodologies, social impact assessment frameworks, or participatory decision-making techniques that would enable more rigorous evaluation of social dimensions.

4.3 Proposed Learning Objectives

To address the identified knowledge gaps, I propose the following specific, achievable learning objectives that leverage my existing strengths in instrumentation engineering and finance whilst developing new competencies aligned with my data science studies:

Objective 1: Within the next three months, develop proficiency in energy system simulation and optimisation using Python, specifically applying machine learning techniques to heat network demand forecasting. Building on my data science coursework, I will complete a personal project that models district heating load profiles using historical consumption data and weather variables. This will bridge my technical instrumentation background with emerging data-driven approaches to energy system management, addressing the gap in dynamic system modelling identified during this case study.

Objective 2: Apply my finance background to develop a comprehensive techno-economic model for waste heat recovery infrastructure within the next four months. Despite holding a Master's degree in finance, I recognise that my previous training focused on financial markets rather than infrastructure project appraisal. I will specifically study levelised cost of heat (LCOH) calculations, public-private partnership structures, and the application of real options analysis to energy infrastructure investments, creating a reusable Excel/Python model template that integrates technical performance parameters with financial metrics.

Objective 3: Enrol in and complete a short course or MOOC on UK energy policy and regulation within the next six months, with particular focus on the Heat Networks (Scotland) Act, the Energy Act 2023 provisions for heat network zoning, and Ofgem's regulatory framework. I will document key regulatory requirements in a personal reference guide and discuss implications with industry professionals through Cranfield's networking events or LinkedIn connections in the energy sector.

These objectives strategically integrate my interdisciplinary background, combining instrumentation engineering's systems thinking, finance's investment appraisal rigour, and data science's analytical capabilities, to develop a more holistic approach to sustainable infrastructure evaluation in future professional practice.

5. Conclusion

This case study has demonstrated that the London Underground waste heat challenge is fundamentally a multidisciplinary problem requiring integration of engineering, economic, environmental, social, legal, and ethical perspectives. The MCDA evaluation identified ventilation shaft heat recovery with district heating distribution as the optimal solution, achieving a weighted score of 7.82 through strong performance in carbon reduction, heat recovery capacity, and fuel poverty alleviation. The Bunhill 2 Energy Centre provides proof of concept for this approach, demonstrating technical feasibility and quantified benefits that support confidence in wider deployment.

The analysis revealed sensitivity to weighting assumptions, with economically-focused evaluation favouring regenerative braking energy recovery. This finding emphasises the importance of explicit value statements in decision-making and the need for transparent justification of evaluation frameworks. Successful implementation of any solution will require continued multi-stakeholder collaboration between Transport for London, local authorities, energy companies, and funding bodies.

Personal reflection has identified specific knowledge gaps in regulatory frameworks and social research methods that constrained the depth of this analysis. Proposed learning objectives provide a structured pathway to address these deficiencies. The group collaboration experience reinforced the value of disciplinary diversity in tackling complex infrastructure challenges, whilst highlighting the importance of structured processes for reconciling divergent perspectives.

References

- Celsius Initiative. (2024). Heat recovery from the London Underground in Islington, United Kingdom. <https://celsiuscity.eu/heat-recovery-from-the-london-underground-in-islington-united-kingdom/>
- CIBSE Journal. (2025). Tunnel vision: The challenge of cooling the London Underground. <https://www.cibsejournal.com/technical/londons-tunnel-vision/>
- Climate Change Committee. (2019). Net Zero: The UK's contribution to stopping global warming. <https://www.theccc.org.uk/publication/net-zero-the-uks-contribution-to-stopping-global-warming/>
- Colloide Engineering. (2023). Colloide launch 'first of its kind' renewable energy project - Bunhill Heat and Power Network. <https://colloide.com/colloide-launch-first-of-its-kind-renewable-energy-project-bunhill-heat-and-power-network/>
- Estévez, R. A., Espinoza, V., Ponce Oliva, R. D., Vásquez-Lavín, F., & Gelcich, S. (2021). Multi-criteria decision analysis for renewable energies: Research trends, gaps and the challenge of improving participation. *Sustainability*, 13(6), 3515. <https://doi.org/10.3390/su13063515>
- GEA. (2021). GEA heat pump upgrades waste heat from the London Underground to warm homes, two leisure centres and a school in Islington. <https://www.gea.com/en/customer-cases/bunhill-2-energy-centre-heats-london-homes-with-gea-heat-pump/>
- GOV.UK. (2018). Heat networks: Ensuring sustained investment and protecting consumers. <https://www.gov.uk/government/publications/heat-networks-developing-a-market-framework/heat-networks-ensuring-sustained-investment-and-protecting-consumers>
- GOV.UK. (2023). Energy Security Bill factsheet: Heat networks regulation and zoning. <https://www.gov.uk/government/publications/energy-security-bill-factsheets/energy-security-bill-factsheet-heat-networks-regulation-and-zoning>
- GOV.UK. (2024). UK heat networks: Market overview. <https://www.gov.uk/government/publications/uk-heat-networks-market-overview/uk-heat-networks-market-overview-accessible-webpage>
- Greater London Authority. (2018). London Environment Strategy. <https://www.london.gov.uk/programmes-and-strategies/environment-and-climate-change/london-environment-strategy>
- Greenham, S. (2023). The impact of heat on London Underground infrastructure in a changing climate. *Weather*, 78(5), 134-140. <https://doi.org/10.1002/wea.4421>
- Hepple, J., Sherrat, J., & Yang, S. (2023). Sustainability and carbon neutrality in UK's district heating: A review and analysis. *Cleaner Energy Systems*, 4, 100045. <https://doi.org/10.1016/j.cles.2023.100045>

- International Railway Journal. (2018). London Underground uses braking energy to power stations. <https://www.railjournal.com/passenger/metros/london-underground-uses-braking-energy-to-power-stations/>
- Islington Council. (2022). World-first scheme is launched using waste heat from the Tube to warm homes, two leisure centres and a school in Islington. <https://www.islington.media/news/bunhill-2-launch-pr>
- Railway Gazette International. (2015). London Underground energy recovery trial proves successful. <https://www.railwaygazette.com/london-underground-energy-recovery-trial-proves-successful/41430.article>
- Rowe, J. M., & Paul, J. D. (2021). Cooling the London Underground: Evaluating the use of groundwater and subterranean river water. *Sustainable Cities and Society*, 75, 103281. <https://doi.org/10.1016/j.scs.2021.103281>
- Transport for London. (2024). London Underground facts. <https://tfl.gov.uk/corporate/about-tfl/what-we-do/london-underground>
- TunnelTalk. (2012). Borehole cooling for Green Park Station. <https://www.tunneltalk.com/London-Underground-Feb12-Borehole-cooling-for-Green-Park-Station.php>
- UK Power Networks Services. (2015). London Underground inverter trial project. <https://www.ukpowernetworksservices.co.uk/case-studies/london-underground-inverter-trial-project/>
- Wang, J. J., Jing, Y. Y., Zhang, C. F., & Zhao, J. H. (2009). Review on multi-criteria decision analysis aid in sustainable energy decision-making. *Renewable and Sustainable Energy Reviews*, 13(9), 2263-2278. <https://doi.org/10.1016/j.rser.2009.06.021>
- Wikipedia. (2025). London Underground cooling. https://en.wikipedia.org/wiki/London_Underground_cooling

Appendix

Percentage Contribution of Heat Sources in London Underground Waste Heat

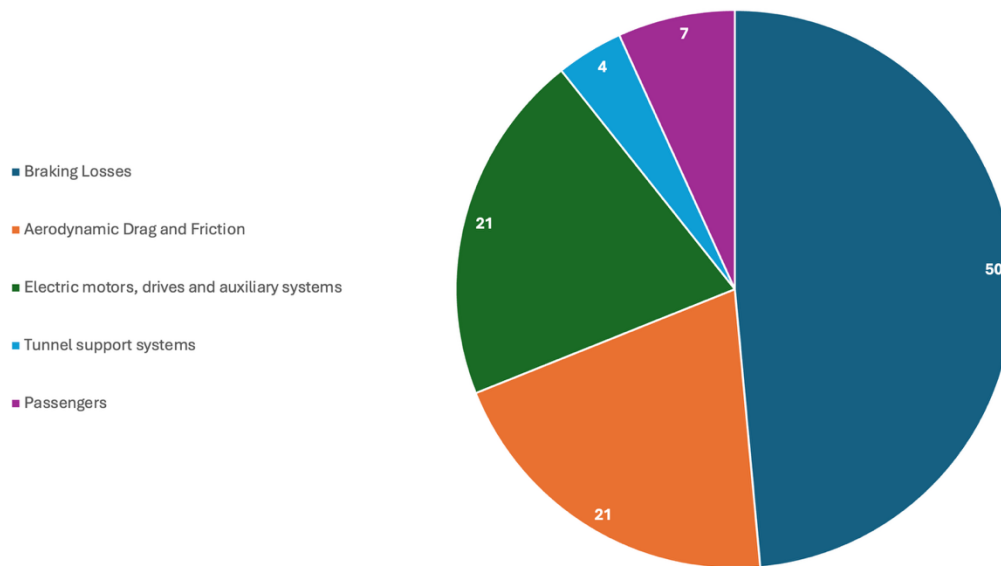


Figure 1

Percentage Contribution of Heat Sources in London Underground

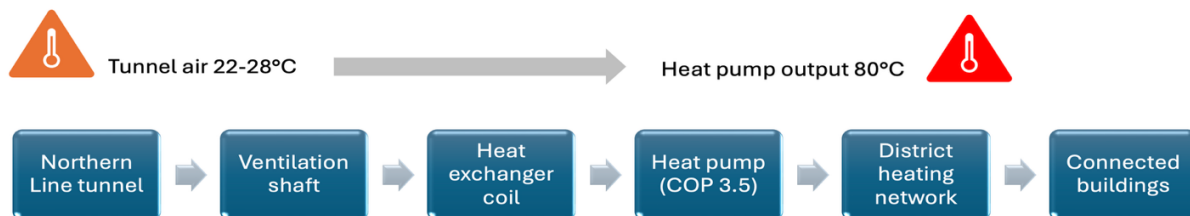


Figure 2

Bunhill 2 Energy Centre System Schematic